Deliverable 6: Data Strategy

What data this needs, how it was simulated, what real data would take

Mills-Operation · AI Reliability Copilot for Hot Mill Operations

Data Strategy

What was needed, and why

Real Nucor sensor data was not available, so it had to be synthesized from scratch. Before writing any generation code, the question was: what is the minimum realistic signal that actually proves a model can see a bearing failure coming before the alarm does, not a generic sensor dataset.

Five signals per roll stand, sampled every minute: vibration (mechanical wear, the standard precursor), bearing temperature (thermal, slower and noisier than vibration), motor current (electrical, ties the fault to something a plant's PLC would already log), line speed (operational response, tests whether the model reads a human throttling back rather than just raw physics), and coolant pressure, the one signal that drops while everything else rises, which forces the model to learn an actual pattern instead of thresholding one direction.

How it was built

The first attempt was a flat baseline with a sudden step change right before failure. It looked fake immediately, flat then a cliff, and no real machine dies like a light switch. Rebuilt as a non linear ramp: starting 6 to 48 hours before failure, each signal drifts toward a failing value on a progress squared curve, slow at first and accelerating near the end, with the noise widening as the fault develops rather than just the mean shifting.

The generator (`data/generate_synthetic_data.py`) is seeded and fully reproducible. Only the script is checked in, not the output. Result: 6 stands times 45 days at 1 minute resolution, about 389K rows, roughly 22% of stand days ending in an injected failure, paired with a ground truth label file. The window anchors to end yesterday off the real clock instead of a fixed date, so the data stays current without ever needing a retrain, same seed and same signal sequences, only the calendar labels move.

This is a best guess at realistic magnitudes and failure dynamics, not something measured off real equipment, and it is worth saying that plainly rather than implying the numbers are more grounded than they are.

What I'd ask Nucor's data team for

- **Historian access** (OSIsoft PI or equivalent) for the real tag names and sampling rates behind these five signals on an actual roll stand. Plant historians often log slower or event triggered, which changes the modeling approach.
- **A real failure log from a CMMS**, actual unplanned downtime with timestamps and root cause, so the model trains against ground truth instead of simulated labels.
- **Reliability engineering sign off on the signal list itself**, whether these five are the real leading indicators for this failure mode at this mill, or whether a sixth, like oil analysis or acoustic emission, matters more in practice.
- **Data quality expectations**: sensor dropout rates, calibration drift, and how stale a reading is allowed to get before it is untrustworthy. Detection is only as good as the data underneath it.